



universität
wien

BACHELORARBEIT / BACHELOR'S THESIS

Titel der Bachelorarbeit / Title of the Bachelor's Thesis

„Taurus velocity Field“

verfasst von / submitted by

Felix Eppacher

angestrebter akademischer Grad / in partial fulfilment of the requirements for the degree of

Bachelor of Science (BSc)

Wien, 2026 / Vienna, 2026

Studienkennzahl lt. Studienblatt /
degree programme code as it appears on
the student record sheet:

UA 033 661

Studienrichtung lt. Studienblatt /
degree programme as it appears on
the student record sheet:

Bachelorstudium Astronomie

Betreut von / Supervisor:

Dr. Sebastian Hutschenreuter, MSc

Abstract

This L^AT_EX template provides example on how to format and display text, mathematical formulas, and insert tables or images. There is a lot more you can do with L^AT_EX, for more information check out <https://en.wikibooks.org/wiki/LaTeX>.

Contents

Abstract	i
1. Introduction	1
2. Taurus Molecular Cloud	3
2.1. Structure and Properties	3
2.2. Kinematics	3
3. Methods	7
3.1. Dataset	7
3.2. Information Field Theory	8
3.3. NIFTy - Numerical Information Field Theory	8
4. Results	11
4.1. The Reconstructed Velocity Field	11
4.2. Taurus Reference Frame	11
4.3. Validation	11
5. Discussion	13
6. Conclusion	15
Bibliography	17
A. Appendix	19

1. Introduction

The local neighborhood of our sun is a region that contains various interesting subjects worth studying. The sun and earth are located approximately in the center of a vast region of relatively empty space containing high-temperature plasma which is surrounded by a shell of interstellar gas and dust. Most star-forming regions near the sun are located on the surface of this shell, which is called the Local Bubble. It is thought to be the result of a burst of supernovae occurring near the center around 14 Myr ago. The likely perpetrators of this event were around 15 young stellar objects in the Scorpius-Centaurus (Sco-Cen) OB association. Sco-Cen has since moved away from the center of the Local Bubble, but the resulting expansion of the shell has continued to push the interstellar medium (ISM) outward and has triggered the creation of new star-forming regions on the surface of the bubble. Due to this property and its close proximity, the local neighborhood is a highly valuable subject for studying star formation and especially the theory of supernova-driven star formation [18].

One of the regions on the surface of the Local Bubble is the Taurus molecular cloud, which is the subject of this thesis. Taurus is one of the most intensively studied star-forming regions in the Milky Way. Its close proximity to the Sun allows young stellar objects (YSOs) to be spatially resolved and characterized individually across the full complex [8]. Unlike denser, high-mass regions such as the Orion Nebula Cluster, Taurus is an example of low-density star formation [11], making it possible to study the formation of solar-type stars in relative isolation. Its stellar population also spans the full range of early evolutionary stages, from deeply embedded protostars to disk-free stars that actively form planets [8], allowing the sequence of stellar evolution to be observed within a single region.

The aim of this thesis is to study the kinematics of the Taurus star-forming region. So far, this has mainly been done by looking at the discrete, scattered velocities of individual stars. However, a full 3D space velocity can only be obtained for stars that also have a measured radial velocity, which is not the case for every member of the complex (only about 1/3), so for many stars only the two-dimensional proper motion is known. Since we don't want this proper motion data to just go to waste, this thesis instead seeks a clearer picture of the region's motions by inferring a three-dimensional velocity field from all the available stellar velocity data. This velocity field is reconstructed using Numerical Information Field Theory (NIFTy), a Bayesian framework for field inference that models the continuous velocity field on a three-dimensional grid. The input data consist of 3D positions, proper motions and, where available, radial velocities all from the latest Gaia DR3 measurements. The resulting velocity field allows us to analyze both the motion of the whole Taurus complex as well as internal kinematics of its substructures.

The thesis is structured as follows: Chapter 2 gives a short introduction to already

1. Introduction

present knowledge about the Taurus Molecular Cloud (TMC) and its kinematics. Chapter 3 briefly explains the methods used to reconstruct a continuous field out of the velocity data. Chapter 4 presents the resulting 3D velocity field and in Chapter 5, we discuss possible findings and caveats of the result.

2. Taurus Molecular Cloud

2.1. Structure and Properties

Taurus lies at the boundary of two expanding interstellar structures, the Local Bubble and the Perseus-Taurus shell. Soler et al. [17] argue that the converging flows driven by both structures triggered its formation. The Orion-Eridanus superbubble also lies in the broader vicinity of Taurus [13], another large, low-density and dust-poor cavity in the local ISM named after the Orion star forming region, which is the assumed source. The proximity to all these large structures (interessante entstehungsmöglichkeiten erwähnen)

An interactive 3D map of the solar neighborhood with the most prominent star-forming regions on the surface of the Local Bubble, as well as the 3D shapes and motions of their young stellar clusters, presented by Zucker et al. [18] can be viewed at ([interactive](#)). A more detailed model, showing the Local Bubble as a system of several shells that are open to the Galactic halo rather than a single fully enclosed shell, was presented by O’Neill et al. [14] and is available at ([interactive](#)); a snapshot of this model is shown in Figure 2.1.

The TMC is one of the most intensively studied low-mass star-forming regions in the Milky Way. It spans a total of about 15×15 degrees on the sky and lies at a distance of approximately 140 parsecs from Earth, although significant depth effects are present along the line of sight, with substructures spanning distances from roughly 128 pc to nearly 200 pc. The complex hosts a few hundred YSOs that are not randomly distributed but instead clustered into small groups [8].

The most recent census, based on Gaia DR3, revised the total membership to 532 adopted members distributed across 13 kinematic groups, most with median ages of 1–3 Myr, while neighbouring associations have ages ranging from 13 to 56 Myr. Samples of older stars (≥ 10 Myr) in the vicinity of Taurus were proposed to also be associated with the cloud but were not included in the catalog because of inconsistent kinematics compared to Taurus [12].

2.2. Kinematics

The kinematics of Taurus have been the subject of several studies, enabled by increasingly precise astrometric data. An early investigation by Bertout and Genova [2] used proper motions supplemented by radial velocities to identify 94 probable members of the Taurus-Auriga moving group and to derive individual kinematic parallaxes, yielding a mean distance of 140 pc and a spatial velocity dispersion of approximately 6 km s^{-1} across the association.

2. Taurus Molecular Cloud

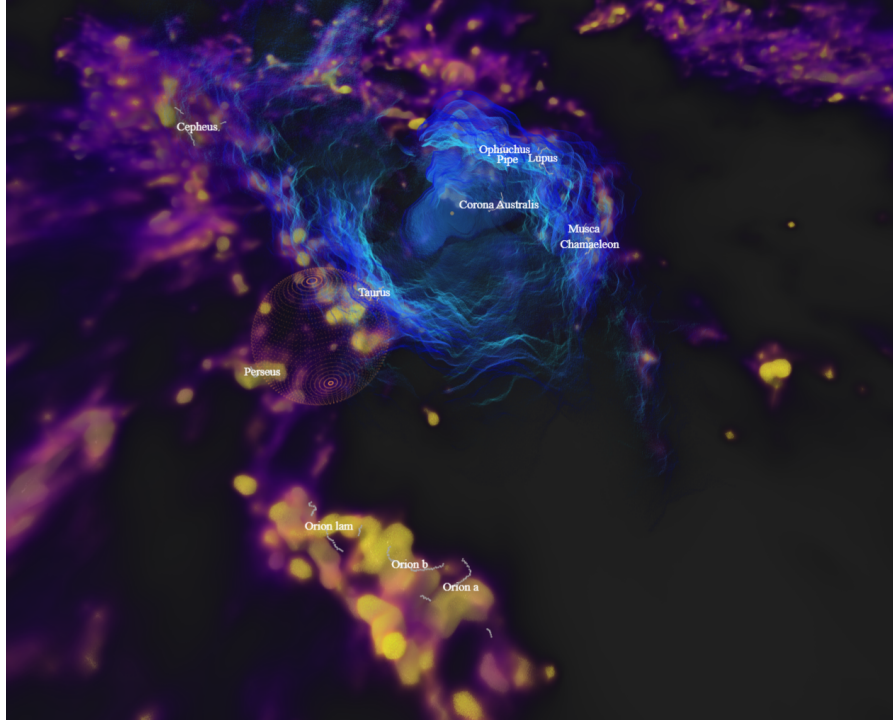


Figure 2.1.: Snapshot of the interactive three-dimensional dust map of the solar neighborhood from O’Neill et al. [14], showing prominent nearby molecular clouds and star-forming regions around the low-density cavity of the Local Bubble; Taurus lies along its edge, adjacent to Perseus.

A more complete and precise picture was established by Galli et al. [8], who combined Gaia DR2 astrometry with VLBI measurements from the GOBELINS project for a sample of 415 stars. Applying a hierarchical clustering algorithm, they identified 21 clusters, of which 13 could be associated with distinct molecular clouds, and derived their distances and spatial velocities. They found inter-cloud relative motions ranging from about 1 to 5 km s^{-1} , a median inter-cloud separation of 25 pc , and a one-dimensional velocity dispersion of the full sample on the order of 2 km s^{-1} . No significant expansion or contraction pattern was detected for the complex as a whole, although evidence for a potential global rotation was reported. The stellar velocities were furthermore found to closely match those of the underlying ^{13}CO molecular gas, confirming a tight dynamical coupling between the stars and the gas.

This picture was extended by Soler et al. [17], who combined 3D dust reconstruction with CO gas emission and neutral hydrogen observations to study the large-scale cloud dynamics. Their analysis revealed a velocity gradient across the cloud consistent with external compression, in line with the bubble geometry described above. Despite this progress, the internal kinematics of Taurus remain incompletely characterized, in part because radial velocity measurements are available for only a subset of its members. This

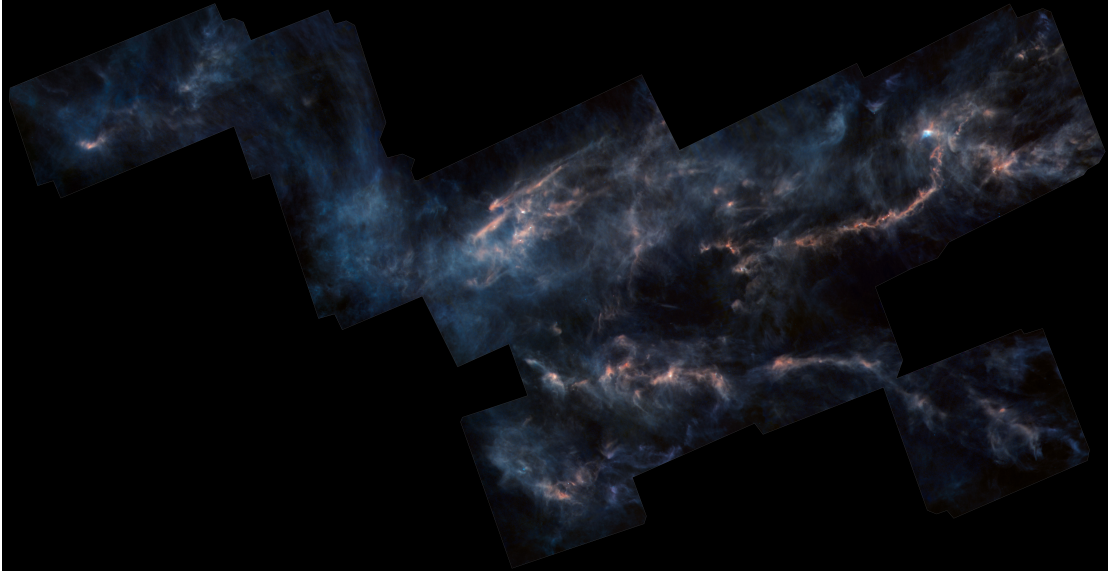


Figure 2.2.: "This mosaic combines several observations of the Taurus Molecular Cloud performed by ESA's Herschel observatory." [6]

motivates a complementary approach that uses the full available data of Gaia proper motions to reconstruct a continuous 3D velocity field across the complex, as pursued in this thesis.

3. Methods

3.1. Dataset

The analysis is based on a catalog of 557 stars associated with the Taurus molecular cloud, combining the membership list of Luhman [12] with the SigMA clustering algorithm of Ratzenböck et al. [15], both derived from Gaia DR3 astrometry. For each star the catalog provides the full astrometric solution: equatorial coordinates (α, δ) , trigonometric parallax ϖ and proper motions in right ascension and declination $(\mu_{\alpha*}, \mu_{\delta})$ together with radial velocities where available, all with their respective uncertainties. As a quality indicator for the astrometric solution serves the renormalized unit weight error (RUWE). All 557 stars possess proper motion measurements, while radial velocities are available for only 180 of them, representing roughly one third of the sample. Photometric data are provided in the form of the Gaia G -band magnitude and the colors $G - G_{\text{RP}}$ and $G_{\text{BP}} - G_{\text{RP}}$. Stellar ages, visual extinctions A_V , and masses, obtained through isochrone fitting with the CHRONOS algorithm [16], are included for a subset of the sample.

The three-dimensional heliocentric Galactic Cartesian positions (X, Y, Z) and corresponding velocity components (U, V, W) are computed from the astrometric and radial-velocity data, with the X -axis pointing toward the Galactic center, Y in the direction of Galactic rotation, and Z toward the Galactic north pole.

Prior to inference, the sample is filtered to remove unreliable measurements. Radial velocities outside the range $[-30, 50] \text{ km s}^{-1}$ are excluded, as well as stars with uncertainties on their RV measurement greater than XXXX. Such cuts were made only for the RV data because it had far greater uncertainties than the proper motions (+ binaries). Due to the limited number of stars with radial-velocity measurements, this range is set deliberately wide in order to retain as many data points as possible, excluding only the most obvious outliers that are clearly inconsistent with membership in Taurus. Stars with $\text{RUWE} > 1.4$ are additionally excluded from the radial-velocity likelihood, as elevated RUWE values are indicative of unresolved binary systems or a poor astrometric solution that may bias the velocity measurement [10]. Here too, a fixed value was chosen above a position-dependent RUWE threshold derived from the Gaia Unlimited binary selection function [7], in order to include as many radial velocities as possible. The resulting catalog spans a spatial volume of roughly $120 \times 75 \times 75 \text{ pc}$ around Taurus, discretized on a grid with a cell size of 3 pc per dimension, matching the resolution at which the velocity field is later reconstructed.

3.2. Information Field Theory

Only a fraction of the stars in the catalog have measured radial velocities, while proper motions, and thus two of the three velocity components, are available for essentially the full sample. Rather than analyzing individual stars or predefined subgroups separately, this thesis combines all of this information into a single, continuous three-dimensional velocity field that describes the motion of Taurus everywhere within the studied volume. However, only at the locations of observed stars does the field yield any meaningful results.

This is a common problem in astrophysics: a continuous physical quantity, here a velocity field, must be reconstructed from a finite, noisy set of measurements. Information field theory (IFT) is a Bayesian framework designed specifically for this kind of problem [4, 5]. Instead of treating the unknown field as a small set of parameters, such as the coefficients of a simple analytic function, IFT treats it as the object to be inferred directly: the field is allowed, in principle, to take on any value at any point in space, and its shape is constrained purely by the data and by a set of reasonable prior assumptions.

The key prior assumption used here is that the velocity field is spatially correlated: nearby points are expected to have similar velocities, while the field can vary more or less smoothly depending on the physical scale being considered. This kind of prior is commonly implemented as a Gaussian process, whose correlation properties are described by a power spectrum encoding how much power, i.e. variation, the field has on large versus small scales. Given this prior and the observed proper motions and radial velocities, Bayes' theorem in principle yields a posterior probability distribution over all possible velocity fields consistent with the data. In practice, this posterior cannot be written down or sampled from exactly, because the number of unknowns, the field value at every grid point, is far too large. IFT provides the mathematical machinery to approximate this posterior numerically, and to do so in a way that also returns uncertainty estimates for the reconstructed field, not just a single best-fit map.

3.3. NIFTy - Numerical Information Field Theory

The numerical implementation of IFT used in this thesis is NIFTy [NIFTy.RE; 3], a Python package built on JAX that extends the original NIFTy framework [1] with ready-to-use building blocks for correlated field models and for the variational inference algorithms needed to approximate the posterior described in Section 3.2.

Concretely, the three-dimensional velocity field is modeled on a regular grid spanning the volume introduced in Section 3.1, with a cell size of 3 pc. Each velocity component is represented as a NIFTy correlated field: a constant offset, corresponding to the expected bulk motion of Taurus as a whole, plus a spatially correlated deviation from that offset whose typical smoothness, i.e. how much small-scale versus large-scale structure is allowed, is itself inferred from the data rather than fixed by hand.

This general modeling strategy, a correlated field with a flexible, data-driven power spectrum, follows and adapts the approach of Hutschenreuter et al. [9], who used the

3.3. NIFTy - Numerical Information Field Theory

same underlying model and code to reconstruct the 3D velocity field of the Scorpius-Centaurus OB association. With a sample of only about 550 stars in Taurus, splitting the inferred motion into multiple separate kinematic components was not expected to be a meaningful or well-constrained distinction; a single correlated field per velocity component is therefore sufficient to capture the bulk motion and its spatial structure across the region.

Before the field itself is inferred, the noise properties of the stellar velocity measurements are estimated directly from the data, since the nominal Gaia uncertainties do not necessarily capture the full observational error budget. Given the noise model and the correlated-field priors, the posterior distribution over the velocity field is then approximated using NIFTy’s variational inference scheme: the algorithm iteratively refines an approximate posterior by alternating between updating the most probable field configuration and drawing an increasing number of posterior samples around it, in order to capture the local shape of the posterior more accurately as the fit converges. The final result of this procedure is a set of posterior samples of the full 3D velocity field, from which both a mean velocity map and a corresponding uncertainty (standard deviation) map are computed at every grid point, providing the reconstructed velocity field of Taurus together with a quantitative estimate of its reliability.

4. Results

4.1. The Reconstructed Velocity Field

A 3D viewer of the resulting vector field and the stellar data is available in ([vector-field viewer](#)).

4.2. Taurus Reference Frame

4.3. Validation

5. Discussion

6. Conclusion

Bibliography

- [1] P. Arras, M. Baltac, T. A. Ensslin, P. Frank, S. Hutschenreuter, J. Knollmueller, R. Leike, M.-N. Newrzella, L. Platz, M. Reinecke, and J. Stadler. NIFTy5: Numerical Information Field Theory v5. Astrophysics Source Code Library, record [ascl:1903.008](#), Mar. 2019.
- [2] C. Bertout and F. Genova. A kinematic study of the Taurus-Auriga T association. *Astronomy & Astrophysics*, 460:499–518, 2006. [doi:10.1051/0004-6361:20065842](#).
- [3] G. Edenhofer, P. Frank, J. Roth, R. H. Leike, M. Guerdi, L. I. Scheel-Platz, M. Guardiani, V. Eberle, M. Westerkamp, and T. A. Enßlin. Re-envisioning numerical information field theory (NIFTy.re): A library for Gaussian processes and variational inference. *Journal of Open Source Software*, 9(98):6593, 2024. [doi:10.21105/joss.06593](#).
- [4] T. A. Enßlin. Information theory & information field theory, 2023. Lecture notes, version March 27, 2023.
- [5] T. A. Enßlin. Information field theory: Concepts, applications, and AI-perspective. 2025.
- [6] ESA/Herschel/NASA/JPL-Caltech. Herschel’s view of the Taurus molecular cloud, Sept. 2017. URL https://www.esa.int/ESA_Multimedia/Images/2017/09/Herschel_s_view_of_the_Taurus_molecular_cloud.
- [7] GaiaUnlimited Collaboration. gaiaunlimited: Selection function tools for Gaia. <https://gaiaunlimited.readthedocs.io>, 2023.
- [8] P. A. B. Galli, L. Loinard, H. Bouy, L. M. Sarro, G. N. Ortiz-León, S. A. Dzib, J. Olivares, M. Heyer, J. Hernandez, C. Román-Zúñiga, M. Kounkel, and K. Covey. Structure and kinematics of the Taurus star-forming region from Gaia-DR2 and VLBI astrometry. *Astronomy & Astrophysics*, 630:A137, 2019. [doi:10.1051/0004-6361/201935928](#).
- [9] S. Hutschenreuter, J. Alves, L. Posch, J. Großschedl, M. Piecka, N. Miret-Roig, S. Ratzenböck, and C. Swiggum. The velocity field of the Scorpius-Centaurus OB association. I. method and general properties. *Astronomy & Astrophysics*, 705:A108, 2026. [doi:10.1051/0004-6361/202557273](#).
- [10] L. Lindegren, S. A. Klioner, J. Hernández, et al. Gaia early data release 3: The astrometric solution. *Astronomy & Astrophysics*, 649:A2, 2021. [doi:10.1051/0004-6361/202039653](#).

Bibliography

- [11] K. L. Luhman. The stellar membership of the Taurus star-forming region. *The Astronomical Journal*, 156:271, 2018. doi:[10.3847/1538-3881/aae831](https://doi.org/10.3847/1538-3881/aae831).
- [12] K. L. Luhman. A census of the Taurus star-forming region and neighboring associations with Gaia. *The Astronomical Journal*, 165(2):37, 2023. doi:[10.3847/1538-3881/ac9da3](https://doi.org/10.3847/1538-3881/ac9da3).
- [13] B. B. Ochsendorf, A. G. A. Brown, J. Bally, and A. G. G. M. Tielens. Nested shells reveal the rejuvenation of the Orion-Eridanus superbubble. *The Astrophysical Journal*, 808(2):111, 2015. doi:[10.1088/0004-637X/808/2/111](https://doi.org/10.1088/0004-637X/808/2/111).
- [14] T. J. O’Neill, C. Zucker, A. A. Goodman, and G. Edenhofer. The Local Bubble Is a Local Chimney: A New Model from 3D Dust Mapping. *The Astrophysical Journal*, 973(2):136, Oct. 2024. doi:[10.3847/1538-4357/ad61de](https://doi.org/10.3847/1538-4357/ad61de).
- [15] S. Ratzenböck, J. E. Großschedl, T. Möller, J. Alves, I. Bomze, and S. Mein-gast. Significance mode analysis (SigMA) for hierarchical structures. An applica-tion to the Sco-Cen OB association. *Astronomy & Astrophysics*, 677:A59, 2023. doi:[10.1051/0004-6361/202243690](https://doi.org/10.1051/0004-6361/202243690).
- [16] Ratzenböck, Sebastian, Großschedl, Josefa E., Alves, João, Miret-Roig, Núria, Bomze, Immanuel, Forbes, John, Goodman, Alyssa, Hacar, Álvaro, Lin, Doug, Mein-gast, Stefan, Möller, Torsten, Piecka, Martin, Posch, Laura, Rottensteiner, Alena, Swiggum, Cameren, and Zucker, Catherine. The star formation history of the sco-cen association - coherent star formation patterns in space and time. *A&A*, 678:A71, 2023. doi:[10.1051/0004-6361/202346901](https://doi.org/10.1051/0004-6361/202346901). URL <https://doi.org/10.1051/0004-6361/202346901>.
- [17] J. D. Soler, C. Zucker, J. E. G. Peek, M. Heyer, P. F. Goldsmith, S. C. O. Glover, S. Molinari, R. S. Klessen, P. Hennebelle, L. Testi, T. Colman, M. Benedettini, D. Elia, C. Mininni, S. Pezzuto, E. Schisano, and A. Traficante. A panoptic view of the Taurus molecular cloud. I. the cloud dynamics revealed by gas emis-sion and 3D dust. *Astronomy & Astrophysics*, 675:A206, 2023. doi:[10.1051/0004-6361/202346241](https://doi.org/10.1051/0004-6361/202346241).
- [18] C. Zucker, A. A. Goodman, J. Alves, S. Bialy, M. Foley, J. S. Speagle, J. Großschedl, D. P. Finkbeiner, A. Burkert, D. Khimey, and C. Swiggum. Star formation near the Sun is driven by expansion of the Local Bubble. *Nature*, 601(7893):334–337, Jan. 2022. doi:[10.1038/s41586-021-04286-5](https://doi.org/10.1038/s41586-021-04286-5).

A. Appendix

here you can put further things you want to add like transcripts, questionnaires, raw data...